YEAR 12 PHYSICS
REVISION

VECTORS

11 A student whirls a rubber stopper on the end of a nylon string in a horizontal clockwise circle (as seen from above) at a constant speed of 6.0 m s^{-1}. From an instant when the stopper is moving in a northerly direction, find its change in velocity after moving round
(*a*) one-half of the circumference;
(*b*) one complete turn;
(*c*) one-quarter of a turn;
(*d*) one-ninth of a turn.

ANS: (a) 12 ms^{-1} S (b) zero
(c) 8.5 ms^{-1} SE (d) 4.1 ms^{-1} E 20° S

19 A yacht travelling with a velocity of 10 km h^{-1} north relative to the water is in a tidal stream flowing south-east at 4.0 km h^{-1}. Determine, either by scale drawing or by calculation, the yacht's velocity relative to the ocean floor.

ANS: 7.8 kmh^{-1} N 22° E

25 A river 2.0 km across has a current of velocity 8.0 km h^{-1}. In what direction must a launch, moving at 16 km h^{-1} relative to the water, steer in order to reach one bank directly opposite its starting point on the other? What will be the time for the crossing?

ANS: 60° to bank upstream, 522 s.

PROJECTILE MOTION

3 A ball is thrown horizontally with a speed of 14.0 m s^{-1} from a point 6.40 m above the ground. Calculate
(*a*) the time taken by the ball to reach ground level;
(*b*) the horizontal distance travelled in that time;
(*c*) its velocity when it reaches the ground.

ANS: (a) 1.14 s (b) 16.0 m (c) 17.9 ms^{-1} at 38.7° to horizontal

8 A stone is thrown with a velocity of 20 m s^{-1} at an angle of 40° with the horizontal. Determine
(*a*) the time for which it is above the level at which it is thrown;
(*b*) the greatest vertical height reached;
(*c*) its range on a horizontal plane.

ANS: (a) 2.6 s (b) 8.4 m (c) 40 m

7 A tennis player hits a lob shot at point X in Figure 13.1 with a velocity of 13 m s^{-1} at an angle of 60° to the horizontal.

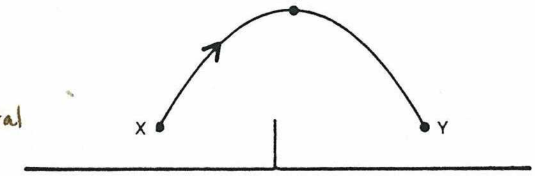


Fig. 13.1

Answer the following questions for the flight of the ball from X to a point Y, the same height above the ground as X.
(*a*) What is the vertical component of the ball's initial velocity?
(*b*) What is the vertical component of the ball's velocity at Z, the highest point of its flight?
(*c*) What is the vertical component of the ball's acceleration as it moves from X to Z?
(*d*) How long does the ball take to travel from X to Z?
(*e*) How long does the ball take to travel from X to Y?
(*f*) What is the horizontal component of the ball's velocity at X?
(*g*) What is the horizontal component of the ball's velocity at Z?
(*h*) What is the horizontal distance from X to Y?

ANS: (a) 11 ms^{-1} (b) zero (c) 9.80 ms^{-2} down (d) 1.1 s
(e) 2.2 s (f) 6.5 ms^{-1} (g) 6.5 ms^{-1} (h) 14 m or 15 m

CIRCULAR MOTION

5 A mass of 5.0 kg is whirled in a horizontal circle at one end of a string 49 cm long, the other end being fixed. If the string when hanging vertically will just support a load of 200 kg mass without breaking, find the maximum whirling speed in revolutions per second.

ANS: 4.5 Hz

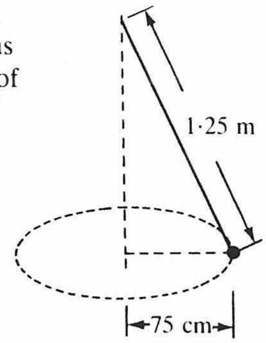
12 A cycling velodrome track is so banked at a certain point, where the horizontal radius of curvature is 30 m, that a cyclist may take the bend at 16 m s^{-1} without depending on a sideways frictional force on his tyres. What is the angle between the horizontal and the track?

ANS: 41°

15 A pilot flies his plane in a vertical loop of radius 500 m at such a speed that at the top of the loop he feels no force from either the seat or the seat belt.
(*a*) At what speed is the plane then flying?
(*b*) If his speed rises to 100 m s^{-1} by the time he reaches the bottom of the loop, what force does he experience from the seat? Take his mass as 67 kg.

ANS: (a) 70.0 ms^{-1} (b) 2.0 x 10^3 N upwards

A body of mass 0·50 kg is hung from a string 1·25 m long and swung around in a horizontal circle of radius 75 cm. Find the tension in the string and the period of revolution of this 'conical pendulum'.

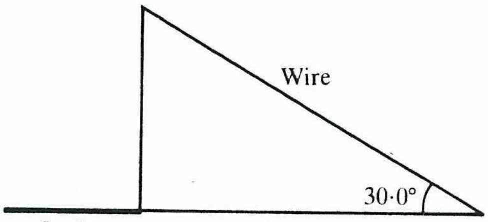


ANS: 6.1 N, 2.0 s

A uniform verandah awning weighing $1{\cdot}00 \times 10^2$ N is supported as shown in the diagram. Find:

(a) the tension in the wire

(b) the reaction at the ledge corner.



ANS: (a) $1{\cdot}00 \times 10^2$ N

(b) $1{\cdot}00 \times 10^2$ N $30{\cdot}0^\circ$ above horizontal

A uniform ladder $1{\cdot}0 \times 10^1$ m long and having a mass of 24 kg rests against a smooth vertical wall with its base on rough level ground. If the base of the ladder is 8·0 m from the wall, find the reaction forces exerted on the ladder by

(a) the wall, and

(b) the ground.

ANS: (a) $1{\cdot}6 \times 10^2$ N

(b) $2{\cdot}8 \times 10^2$ N $56{\cdot}2^\circ$ above horizontal

STRESS AND STRAIN

2 A load of 16.3 kg causes an extension of 3.84 mm in a wire of cross-sectional area 1.0 mm^2 and 6.0 m long.

(*a*) What is the normal (or linear) stress on the wire?

(*b*) What is the normal (or linear) strain?

(*c*) Determine the Young modulus for the material of which the wire is made.

ANS: (a) $1{\cdot}6 \times 10^8$ Pa (b) $6{\cdot}4 \times 10^{-4}$ (c) $2{\cdot}5 \times 10^{11}$ Pa

7 The *ultimate strength* (also known as the *ultimate tensile stress* or *breaking stress*) of a material is—not surprisingly—the maximum stress it can withstand before breaking. The ultimate strength of the steel used for a particular car towing cable is 7.0×10^8 Pa and the Young modulus for the material is 2.0×10^{11} Pa. If the cable has a length of 4.0 m and a total cross-sectional area of 0.20 cm^2,

(*a*) what maximum tension can it withstand?

(*b*) what is the maximum extension possible without breaking, if we assume for simplicity that elastic behaviour continues right up to the instant when fracture occurs?

ANS: (a) $1{\cdot}4 \times 10^4$ N (b) 14 mm

4 Figure 27.1 shows graphically the result of an experiment in which an aluminium wire of diameter 0.50 mm and length 2.5 m was stretched.

(*a*) What is the Young modulus for aluminium?

(*b*) What is the elastic limit for the aluminium wire?

(*c*) As the wire reached its elastic limit,

(*i*) what weight was acting on it?

(*ii*) what was its extension?

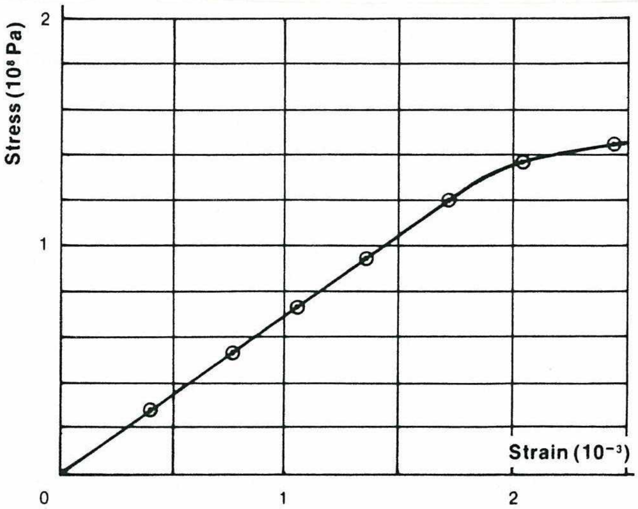


Fig. 27.1

ANS: (a) $(6{\cdot}8 - 7{\cdot}2) \times 10^{10}$ Pa (b) $(1{\cdot}25 - 1{\cdot}35) \times 10^8$ Pa

(c) (i) 25 - 27 N

(ii) 4·5 - 4·7 mm

GRAVITATION

Object	Mass (kg)	Radius (m)	Period of rotation (s)	Mean radius of orbit (m)	(AU)*	Period of revolution (s)	(y)*
Sun	1.98×10^{30}	6.95×10^{8}	2.14×10^{6}				
Mercury	3.28×10^{23}	2.57×10^{6}	5.05×10^{6}	5.79×10^{10}	0.387	7.60×10^{6}	0.241
Venus	4.83×10^{24}	6.31×10^{6}	2.10×10^{7}	1.08×10^{11}	0.723	1.94×10^{7}	0.615
Earth	5.98×10^{24}	6.38×10^{6}	8.61×10^{4}	1.49×10^{11}	1.00	3.16×10^{7}	1.00
Mars	6.37×10^{23}	3.43×10^{6}	8.85×10^{4}	2.28×10^{11}	1.52	5.94×10^{7}	1.88
Jupiter	1.90×10^{27}	7.18×10^{7}	3.54×10^{4}	7.78×10^{11}	5.20	3.74×10^{8}	11.86
Saturn	5.67×10^{26}	6.03×10^{7}	3.60×10^{4}	1.43×10^{12}	9.54	9.30×10^{8}	29.46
Uranus	8.80×10^{25}	2.67×10^{7}	3.88×10^{4}	2.87×10^{12}	19.19	2.66×10^{9}	84.01
Neptune	1.03×10^{26}	2.48×10^{7}	5.69×10^{4}	4.50×10^{12}	30.07	5.20×10^{9}	164.78
Pluto	5.37×10^{23}	2.96×10^{6}	5.51×10^{5}	5.91×10^{12}	39.52	7.82×10^{9}	248.42
Moon	7.34×10^{22}	1.74×10^{6}	2.36×10^{6}	3.84×10^{8}		2.36×10^{6}	

14 How far were the Apollo spacecrafts from the centre of the Earth when they experienced a zero net gravitational force from the Earth and the Moon?

ANS: 3.48×10^{8} m

20 Callisto, Jupiter's largest moon, completes an orbit every 16 d 16 h 32 min at a mean distance of 1.88×10^{9} m from the centre of Jupiter. Deduce a value for the mass of Jupiter.

ANS: 1.89×10^{27} kg

16 A planet which has a diameter twice that of the Earth has a mass six times greater.
(*a*) What is the ratio of the gravitational field at the surface of this planet to that at the surface of the Earth?
(*b*) What is the ratio of the density of the planet to that of the Earth?

ANS: (a) 1·5 (b) 0·75

A moon of Saturn moves in circular orbit of radius $1{\cdot}00 \times 10^{9}$ m and period $1{\cdot}00 \times 10^{6}$ s. Calculate:
(a) the speed at which the moon orbits the planet, and
(b) the mass of Saturn.

ANS: (a) $6{\cdot}28 \times 10^{3}$ ms^{-1} (b) $6{\cdot}0 \times 10^{26}$ kg

MOMENTS, EQUILIBRIUM

A metre rule is pivoted at its mid-point. A 1·2 N force and a 2·4 N force are acting downwards at the 30·0 cm mark and the 90·0 cm mark respectively. Where must an upward force of 3·6 N be applied to balance the rule?

ANS: at 70 cm mark

A uniform bar 4·00 m long is suspended by two vertical ropes, one attached to each end. If the bar has a weight of 19·6 N and has weights of 49·0 N and 98·0 N situated at points 1·00 m and 3·00 m respectively from one end, find the tension in each rope.

ANS: 71·0N, 95·6N

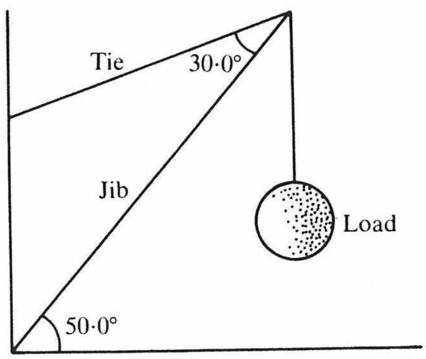
A uniform 4·00 m plank with a mass of 54·0 kg is supported at its ends by two ladders, and two 90·0 kg men stand on it. One man is 1·00 m from one end and the other is 1·50 m from the other end. Determine the forces exerted by the ladders.

ANS: $1{\cdot}04 \times 10^{3}$ N, $1{\cdot}26 \times 10^{3}$ N

A uniform drawbridge of mass $2{\cdot}4 \times 10^{3}$ kg is 7·2 m long. It is pivoted at one end at the bottom of a wall, and it can be raised by a chain which is connected to the drawbridge 4·8 m from the pivot and to the wall 4·8 m above the pivot. When the tension in the chain is just sufficient to start raising the drawbridge from the horizontal position, calculate:
(a) the tension in the chain
(b) the total force exerted on the pivot.

ANS: (a) $2{\cdot}5 \times 10^{4}$ N
(b) $1{\cdot}9 \times 10^{4}$ N 18·4° above horizontal

The diagram represents a jib crane. What are the forces in the jib and tie when the load is $8{\cdot}00 \times 10^{2}$ kg?



ANS: Jib: $1{\cdot}47 \times 10^{4}$ N
Tie: $1{\cdot}01 \times 10^{4}$ N

SECTION C : Comprehension and Interpretation

Marks Allotted: 40 marks out of 200 marks total (20%)

BOTH questions should be attempted.

Read both passages carefully and answer all of the questions at the end of each passage. Candidates are reminded of the need for correct English and clear and concise presentation of answers. Diagrams (sketches), equations and/or numerical results should be included where appropriate.

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1. THE PHYSICS OF THE SQUASH RACQUET [20 marks total]

(Paragraph 1)
The design and construction of squash racquets has undergone several changes over recent years from the round-headed wooden racquet with either a wooden or steel shaft to the oversized head, often of non-circular shape and made from aluminium alloy or composites containing glass, carbon, boron or kevlar fibre. The differing dynamic responses of these materials now gives the user much more scope in choosing a racquet which suits, or sometimes does not suit, their game. We can use some simple physics to help define the criteria for designing and choosing such racquets.

(Paragraph 2)
Most readers will be familiar with the concept of the centre of percussion or "sweet spot" on a bat or racquet. You can easily find the centre of percussion of your racquet by holding it between thumb and forefinger at the position where you would normally grip it and measure the period of oscillation. Using this period in the formula for a simple pendulum

$$T = 2\pi \sqrt{\frac{l}{g}}$$

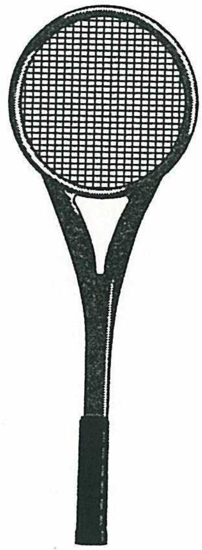
and solving for ℓ gives the distance from the position being held to the centre of percussion. For my racquet I obtained 49 cm, which put the sweet spot right in the middle of the head area.

(Paragraph 3)
The significance of this point is that if we consider (for ease of discussion) a ball striking a stationary racquet at the centre of percussion, then the combination of the translation of the centre of mass backwards and rotation of the racquet about the centre of mass produces one point on the racquet which doesn't move - the point where you are holding it. Hence there is no reaction on the user's arm.

(Paragraph 4)
However, this consideration makes it clear that the centre of percussion and the hand position are conjugate points - so if you don't hit the ball in the sweet spot then the point of zero reaction will be somewhat different from where you are holding it. Then it jars! For the racquet designer this produces a problem because a significant number of players hold the racquet where the grip joins the shaft. Why do they do this? The answer is that the racquet is not a rigid body but can flex roughly about the midpoint of the shaft. Since it is not tied at the ends like a guitar string then the nodes of the fundamental mode of vibration are moved in from the ends with one being at the top of the handle. Thus, gripping the racquet at this point reduces the effect on the user's arm of the oscillation of the racquet.

(Paragraph 5)
This gives two possible causes for sore arms from racquet sports - one due to impact (jarring) and the other to oscillation, which will be worst if the oscillation happens to strike a sympathetic resonance in part of your arm structure. The vibrational frequencies of squash racquets can vary from about 100-200 Hz so a racquet which may suit you may be painful for a friend with a different build. The designer can make the amplitude smaller by making the frame stiffer, but this increases the impact shock. The compromise design will also include good damping so that the oscillations die away quickly. Typically the injection moulded composites have the best damping and aluminium the worst while the compression moulded composites have the highest frequencies.

(a) Paragraph 2 describes how to determine the position of the centre of percussion of a racquet. On following the instructions you find that the time for ten oscillations of your racquet is 12.3 s. Where is the centre of percussion of your racquet? [5 marks]



(b) If you do not hit the ball at the centre of percussion, you "jar" your arm. Describe why. [6 marks]

(c) The article states that "... sore arms ... due to ... oscillation (of the racquet), which will be worst if the oscillation happens to strike a sympathetic resonance in part of your arm structure". What is sympathetic resonance and why will soreness be worst as described? [5 marks]

(d) The vibrational frequencies of squash racquets vary from about 100 to 200 Hz. State two factors which might determine the vibrational frequency of a racquet. [4 marks]

First factor :

Second factor :